

Project / Task

MATRICES / NUMPY ----- Matrix is the tabular representation of the data

Lot of datas are stored in table format,that is why Matrices is very very important topic in python

as we working on dataframe so matrices are played a major role

List is one dimension & matrix is multidimension

indexation is very important to plot the datapoints

```
In [16]: import numpy as np

#Seasons
Seasons = ["2010", "2011", "2012", "2013", "2014", "2015", "2016", "2017", "2018", "2019"]
Sdict = {"2010":0, "2011":1, "2012":2, "2013":3, "2014":4, "2015":5, "2016":6, "2017":7

#Players
Players = ["Sachin", "Rahul", "Smith", "Sami", "Pollard", "Morris", "Samson", "Dhoni", "
Pdct = {"Sachin":0, "Rahul":1, "Smith":2, "Sami":3, "Pollard":4, "Morris":5, "Samson"

#Salaries
Sachin_Salary = [15946875, 17718750, 19490625, 21262500, 23034375, 24806250, 25244493,
Rahul_Salary = [12000000, 12744189, 13488377, 14232567, 14976754, 16324500, 18038573, 1
Smith_Salary = [4621800, 5828090, 13041250, 14410581, 15779912, 14500000, 16022500, 175
Sami_Salary = [3713640, 4694041, 13041250, 14410581, 15779912, 17149243, 18518574, 1945
Pollard_Salary = [4493160, 4806720, 6061274, 13758000, 15202590, 16647180, 18091770, 19
Morris_Salary = [3348000, 4235220, 12455000, 14410581, 15779912, 14500000, 16022500, 17
Samson_Salary = [3144240, 3380160, 3615960, 4574189, 13520500, 14940153, 16359805, 1777
Dhoni_Salary = [0, 0, 4171200, 4484040, 4796880, 6053663, 15506632, 16669630, 17832627, 1
Kohli_Salary = [0, 0, 0, 4822800, 5184480, 5546160, 6993708, 16402500, 17632688, 18862875
Sky_Salary = [3031920, 3841443, 13041250, 14410581, 15779912, 14200000, 15691000, 17182

#Matrix
Salary = np.array([Sachin_Salary, Rahul_Salary, Smith_Salary, Sami_Salary, Polla

#Games
Sachin_G = [80, 77, 82, 82, 73, 82, 58, 78, 6, 35]
Rahul_G = [82, 57, 82, 79, 76, 72, 60, 72, 79, 80]
Smith_G = [79, 78, 75, 81, 76, 79, 62, 76, 77, 69]
Sami_G = [80, 65, 77, 66, 69, 77, 55, 67, 77, 40]
Pollard_G = [82, 82, 82, 79, 82, 78, 54, 76, 71, 41]
Morris_G = [70, 69, 67, 77, 70, 77, 57, 74, 79, 44]
Samson_G = [78, 64, 80, 78, 45, 80, 60, 70, 62, 82]
Dhoni_G = [35, 35, 80, 74, 82, 78, 66, 81, 81, 27]
Kohli_G = [40, 40, 40, 81, 78, 81, 39, 0, 10, 51]
Sky_G = [75, 51, 51, 79, 77, 76, 49, 69, 54, 62]

#Matrix
Games = np.array([Sachin_G, Rahul_G, Smith_G, Sami_G, Pollard_G, Morris_G, Samson_G, Dhoni_G, Kohli_G, Sky_G])

#Points
Sachin_PTS = [2832, 2430, 2323, 2201, 1970, 2078, 1616, 2133, 83, 782]
Rahul_PTS = [1653, 1426, 1779, 1688, 1619, 1312, 1129, 1170, 1245, 1154]
Smith_PTS = [2478, 2132, 2250, 2304, 2258, 2111, 1683, 2036, 2089, 1743]
```

```

Sami_PTS = [2122,1881,1978,1504,1943,1970,1245,1920,2112,966]
Pollard_PTS = [1292,1443,1695,1624,1503,1784,1113,1296,1297,646]
Morris_PTS = [1572,1561,1496,1746,1678,1438,1025,1232,1281,928]
Samson_PTS = [1258,1104,1684,1781,841,1268,1189,1186,1185,1564]
Dhoni_PTS = [903,903,1624,1871,2472,2161,1850,2280,2593,686]
Kohli_PTS = [597,597,597,1361,1619,2026,852,0,159,904]
Sky_PTS = [2040,1397,1254,2386,2045,1941,1082,1463,1028,1331]
#Matrix
Points = np.array([Sachin_PTS, Rahul_PTS, Smith_PTS, Sami_PTS, Pollard_PTS, Morr

```

In [17]: Seasons

```

Out[17]: ['2010',
          '2011',
          '2012',
          '2013',
          '2014',
          '2015',
          '2016',
          '2017',
          '2018',
          '2019']

```

In [18]: Games

```

Out[18]: array([[80, 77, 82, 82, 73, 82, 58, 78, 6, 35],
                [82, 57, 82, 79, 76, 72, 60, 72, 79, 80],
                [79, 78, 75, 81, 76, 79, 62, 76, 77, 69],
                [80, 65, 77, 66, 69, 77, 55, 67, 77, 40],
                [82, 82, 82, 79, 82, 78, 54, 76, 71, 41],
                [70, 69, 67, 77, 70, 77, 57, 74, 79, 44],
                [78, 64, 80, 78, 45, 80, 60, 70, 62, 82],
                [35, 35, 80, 74, 82, 78, 66, 81, 81, 27],
                [40, 40, 40, 81, 78, 81, 39, 0, 10, 51],
                [75, 51, 51, 79, 77, 76, 49, 69, 54, 62]])

```

In [19]: Points

```

Out[19]: array([[2832, 2430, 2323, 2201, 1970, 2078, 1616, 2133, 83, 782],
                [1653, 1426, 1779, 1688, 1619, 1312, 1129, 1170, 1245, 1154],
                [2478, 2132, 2250, 2304, 2258, 2111, 1683, 2036, 2089, 1743],
                [2122, 1881, 1978, 1504, 1943, 1970, 1245, 1920, 2112, 966],
                [1292, 1443, 1695, 1624, 1503, 1784, 1113, 1296, 1297, 646],
                [1572, 1561, 1496, 1746, 1678, 1438, 1025, 1232, 1281, 928],
                [1258, 1104, 1684, 1781, 841, 1268, 1189, 1186, 1185, 1564],
                [ 903,  903, 1624, 1871, 2472, 2161, 1850, 2280, 2593, 686],
                [ 597,  597,  597, 1361, 1619, 2026, 852,  0, 159, 904],
                [2040, 1397, 1254, 2386, 2045, 1941, 1082, 1463, 1028, 1331]])

```

In [20]: Salary

```
Out[20]: array([[15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
                25244493, 27849149, 30453805, 23500000],
               [12000000, 12744189, 13488377, 14232567, 14976754, 16324500,
                18038573, 19752645, 21466718, 23180790],
               [ 4621800,  5828090, 13041250, 14410581, 15779912, 14500000,
                16022500, 17545000, 19067500, 20644400],
               [ 3713640,  4694041, 13041250, 14410581, 15779912, 17149243,
                18518574, 19450000, 22407474, 22458000],
               [ 4493160,  4806720,  6061274, 13758000, 15202590, 16647180,
                18091770, 19536360, 20513178, 21436271],
               [ 3348000,  4235220, 12455000, 14410581, 15779912, 14500000,
                16022500, 17545000, 19067500, 20644400],
               [ 3144240,  3380160,  3615960,  4574189, 13520500, 14940153,
                16359805, 17779458, 18668431, 20068563],
               [      0,      0,  4171200,  4484040,  4796880,  6053663,
                15506632, 16669630, 17832627, 18995624],
               [      0,      0,      0,  4822800,  5184480,  5546160,
                6993708, 16402500, 17632688, 18862875],
               [ 3031920,  3841443, 13041250, 14410581, 15779912, 14200000,
                15691000, 17182000, 18673000, 15000000]])
```

```
In [21]: Pdict
```

```
Out[21]: {'Sachin': 0,
          'Rahul': 1,
          'Smith': 2,
          'Sami': 3,
          'Pollard': 4,
          'Morris': 5,
          'Samson': 6,
          'Dhoni': 7,
          'Kohli': 8,
          'Sky': 9}
```

```
In [22]: Salary/Games
```

```

Out[22]: array([[ 199335.9375      ,  230113.63636364,  237690.54878049,
  259298.7804878   ,  315539.38356164,  302515.24390244,
  435249.87931034,  357040.37179487,  5075634.16666667,
  671428.57142857],
 [ 146341.46341463,  223582.26315789,  164492.40243902,
  180159.07594937,  197062.55263158,  226729.16666667,
  300642.88333333,  274342.29166667,  271730.60759494,
  289759.875       ],
 [  58503.79746835,   74719.1025641  ,  173883.33333333,
  177908.40740741,  207630.42105263,  183544.30379747,
  258427.41935484,  230855.26315789,  247629.87012987,
  299194.20289855],
 [  46420.5        ,   72216.01538462,  169366.88311688,
  218342.13636364,  228694.37681159,  222717.44155844,
  336701.34545455,  290298.50746269,  291006.15584416,
  561450.          ],
 [  54794.63414634,   58618.53658537,   73917.97560976,
  174151.89873418,  185397.43902439,  213425.38461538,
  335032.77777778,  257057.36842105,  288918.          ,
  522835.87804878],
 [  47828.57142857,   61380.          ,  185895.52238806,
  187150.4025974  ,  225427.31428571,  188311.68831169,
  281096.49122807,  237094.59459459,  241360.75949367,
  469190.90909091],
 [  40310.76923077,   52815.          ,   45199.5        ,
   58643.44871795,  300455.55555556,  186751.9125     ,
  272663.41666667,  253992.25714286,  301103.72580645,
  244738.57317073],
 [    0.          ,    0.          ,   52140.          ,
  60595.13513514,  58498.53658537,  77611.06410256,
  234948.96969697,  205797.90123457,  220155.88888889,
  703541.62962963],
 [    0.          ,    0.          ,    0.          ,
  59540.74074074,  66467.69230769,  68471.11111111,
  179325.84615385,          inf, 1763268.8        ,
  369860.29411765],
 [  40425.6        ,   75322.41176471,  255710.78431373,
  182412.41772152,  204933.92207792,  186842.10526316,
  320224.48979592,  249014.49275362,  345796.2962963  ,
  241935.48387097]])

```

```
In [23]: Salary//Games
```

```
Out[23]: array([[ 199335,  230113,  237690,  259298,  315539,  302515,  435249,
                  357040,  5075634,  671428],
                [ 146341,  223582,  164492,  180159,  197062,  226729,  300642,
                  274342,  271730,  289759],
                [  58503,   74719,  173883,  177908,  207630,  183544,  258427,
                  230855,  247629,  299194],
                [  46420,   72216,  169366,  218342,  228694,  222717,  336701,
                  290298,  291006,  561450],
                [  54794,   58618,   73917,  174151,  185397,  213425,  335032,
                  257057,  288918,  522835],
                [  47828,   61380,  185895,  187150,  225427,  188311,  281096,
                  237094,  241360,  469190],
                [  40310,   52815,   45199,   58643,  300455,  186751,  272663,
                  253992,  301103,  244738],
                [     0,         0,   52140,   60595,   58498,   77611,  234948,
                  205797,  220155,  703541],
                [     0,         0,         0,   59540,   66467,   68471,  179325,
                   0, 1763268,  369860],
                [  40425,   75322,  255710,  182412,  204933,  186842,  320224,
                  249014,  345796,  241935]])
```

```
In [24]: np.round(Salary//Games)
```

```
Out[24]: array([[ 199335,  230113,  237690,  259298,  315539,  302515,  435249,
                  357040,  5075634,  671428],
                [ 146341,  223582,  164492,  180159,  197062,  226729,  300642,
                  274342,  271730,  289759],
                [  58503,   74719,  173883,  177908,  207630,  183544,  258427,
                  230855,  247629,  299194],
                [  46420,   72216,  169366,  218342,  228694,  222717,  336701,
                  290298,  291006,  561450],
                [  54794,   58618,   73917,  174151,  185397,  213425,  335032,
                  257057,  288918,  522835],
                [  47828,   61380,  185895,  187150,  225427,  188311,  281096,
                  237094,  241360,  469190],
                [  40310,   52815,   45199,   58643,  300455,  186751,  272663,
                  253992,  301103,  244738],
                [     0,         0,   52140,   60595,   58498,   77611,  234948,
                  205797,  220155,  703541],
                [     0,         0,         0,   59540,   66467,   68471,  179325,
                   0, 1763268,  369860],
                [  40425,   75322,  255710,  182412,  204933,  186842,  320224,
                  249014,  345796,  241935]])
```

```
In [25]: import warnings
warnings.filterwarnings('ignore')
```

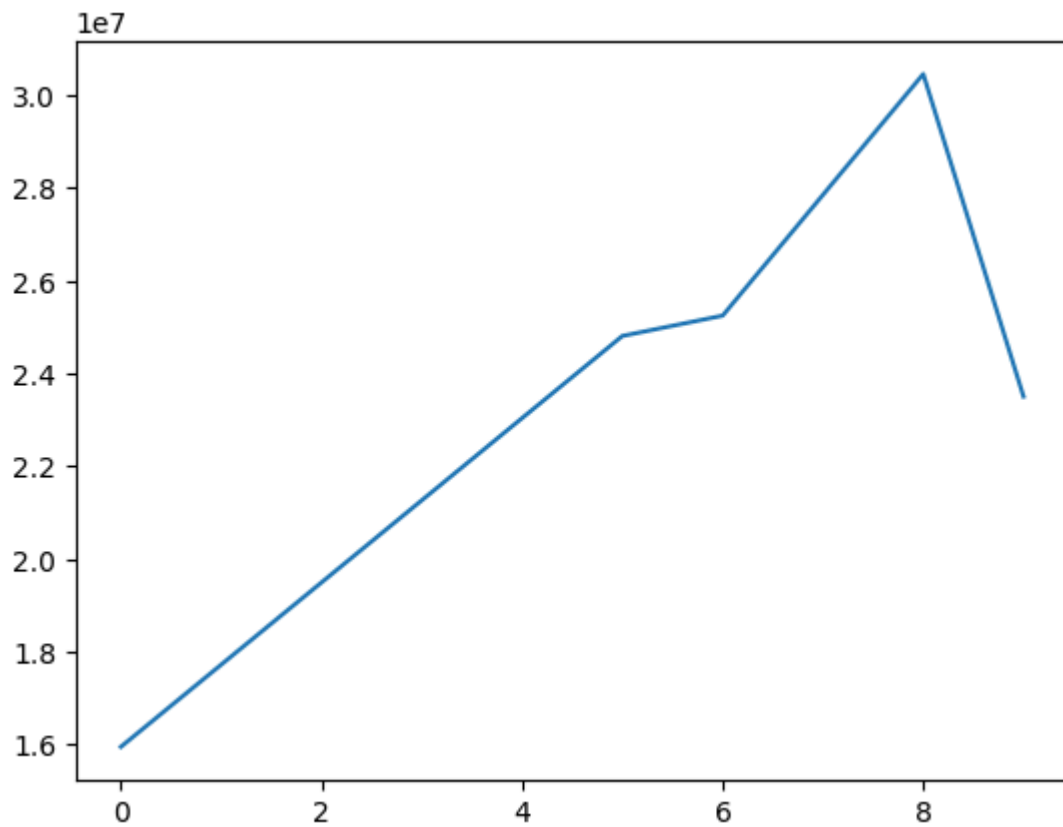
```
In [26]: import matplotlib.pyplot as plt # Libraray used for visualization
```

```
In [27]: Salary[0]
```

```
Out[27]: array([15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
                25244493, 27849149, 30453805, 23500000])
```

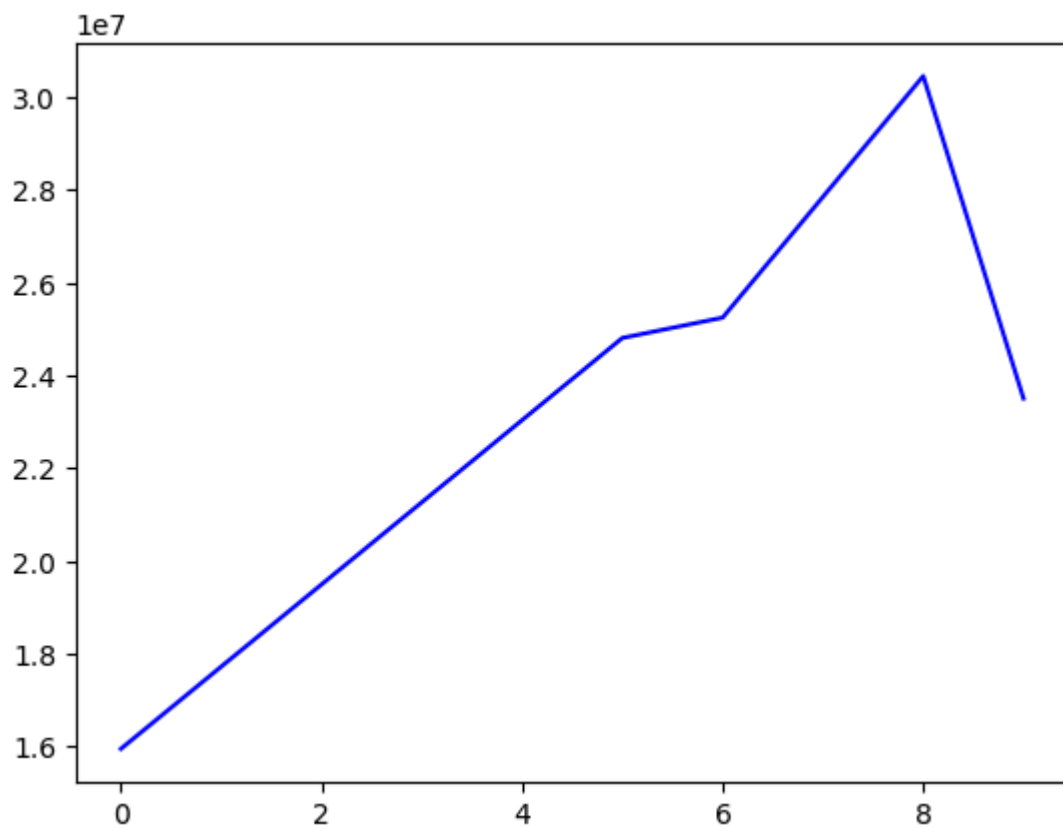
```
In [28]: plt.plot(Salary[0])
```

```
Out[28]: [<matplotlib.lines.Line2D at 0x272cf7e39d0>]
```



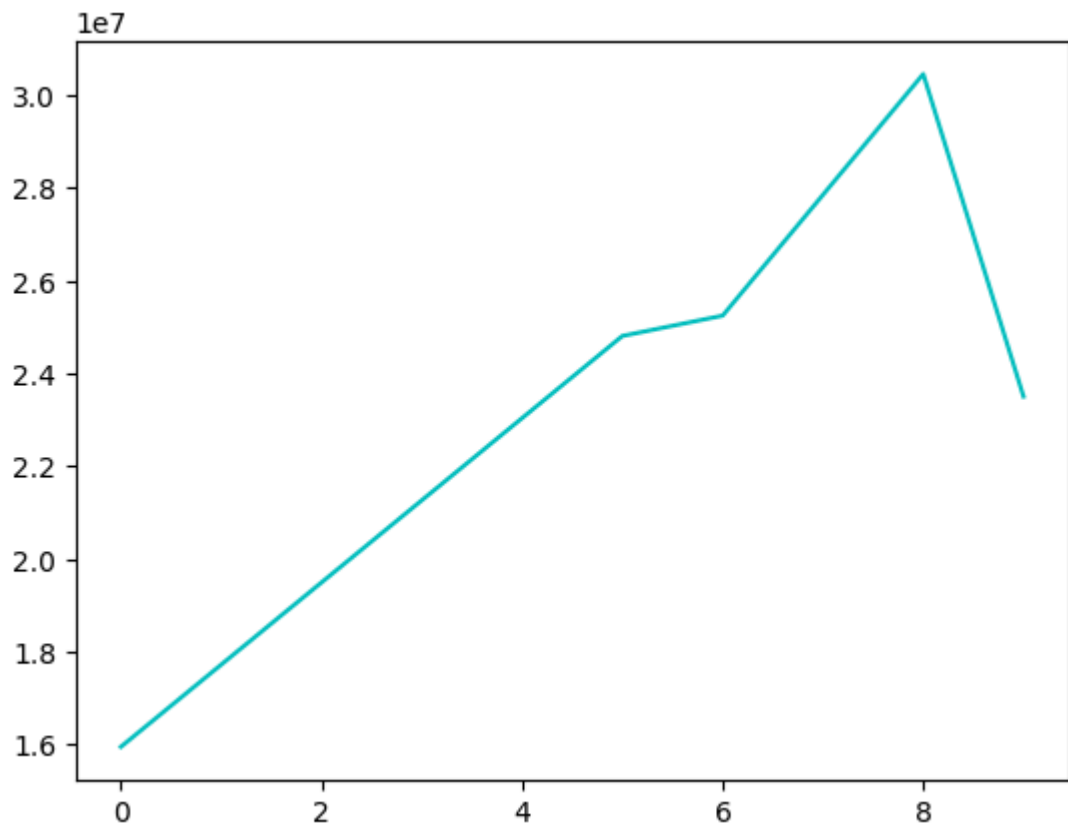
```
In [29]: plt.plot(Salary[0], c = 'b')
```

```
Out[29]: [<matplotlib.lines.Line2D at 0x272cf86df90>]
```



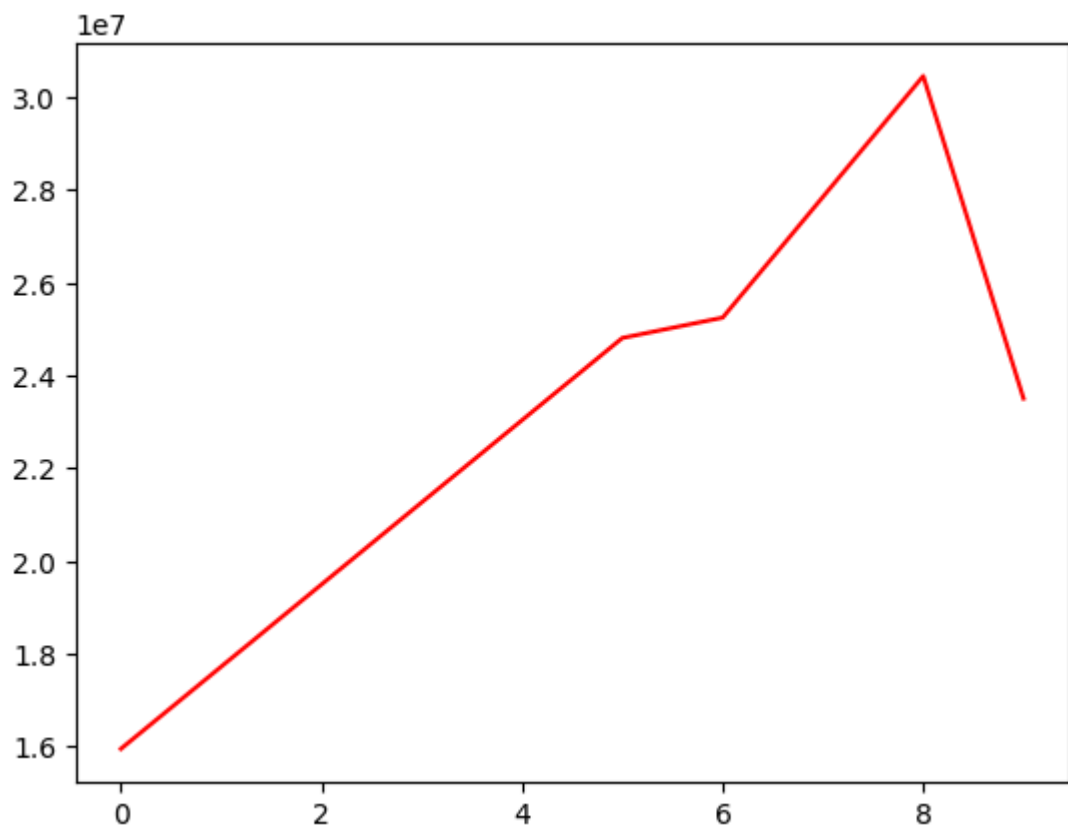
```
In [30]: plt.plot(Salary[0], c = 'r' )
```

```
Out[30]: [<matplotlib.lines.Line2D at 0x272cf6b4550>]
```



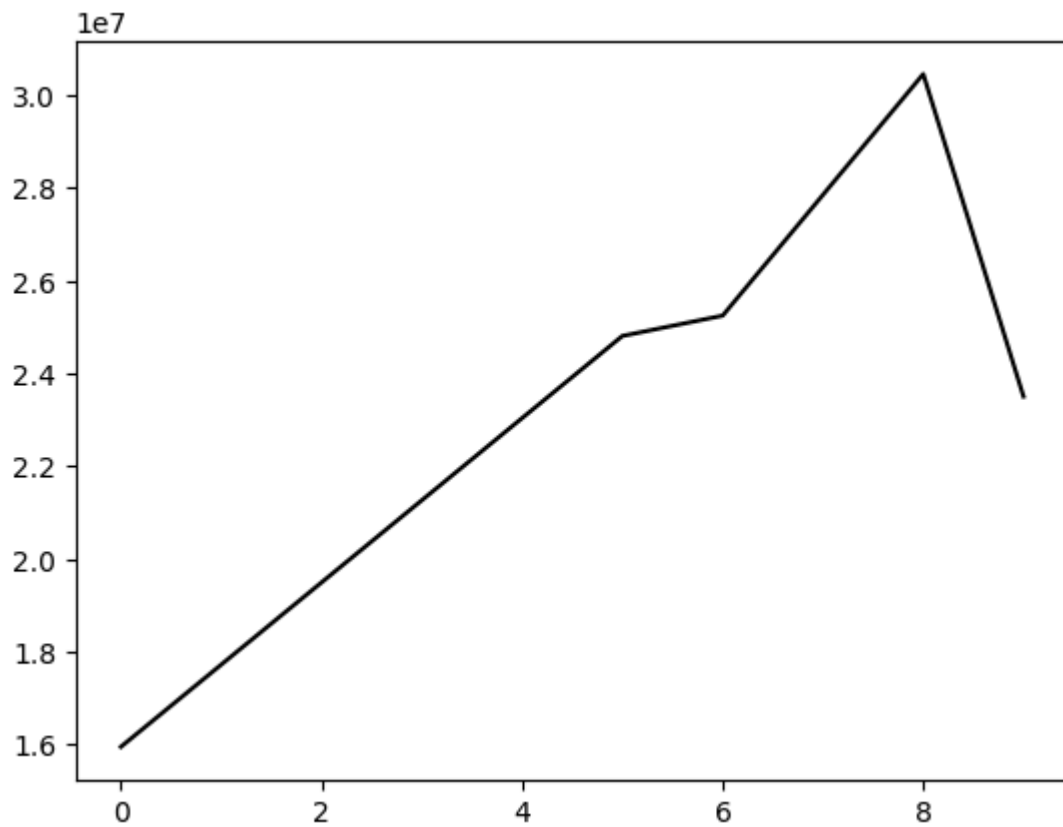
```
In [31]: plt.plot(Salary[0], c='r')
```

```
Out[31]: [<matplotlib.lines.Line2D at 0x272cf712ad0>]
```



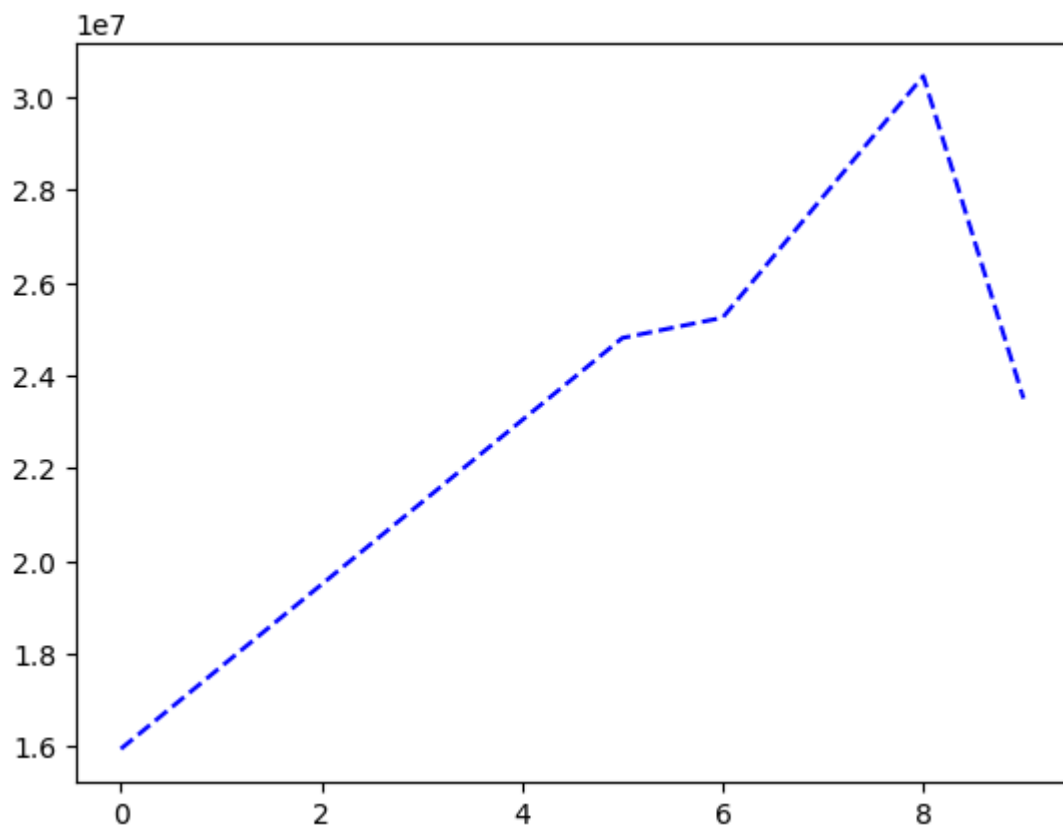
```
In [32]: plt.plot(Salary[0], c='k')
```

```
Out[32]: [<matplotlib.lines.Line2D at 0x272cf759090>]
```



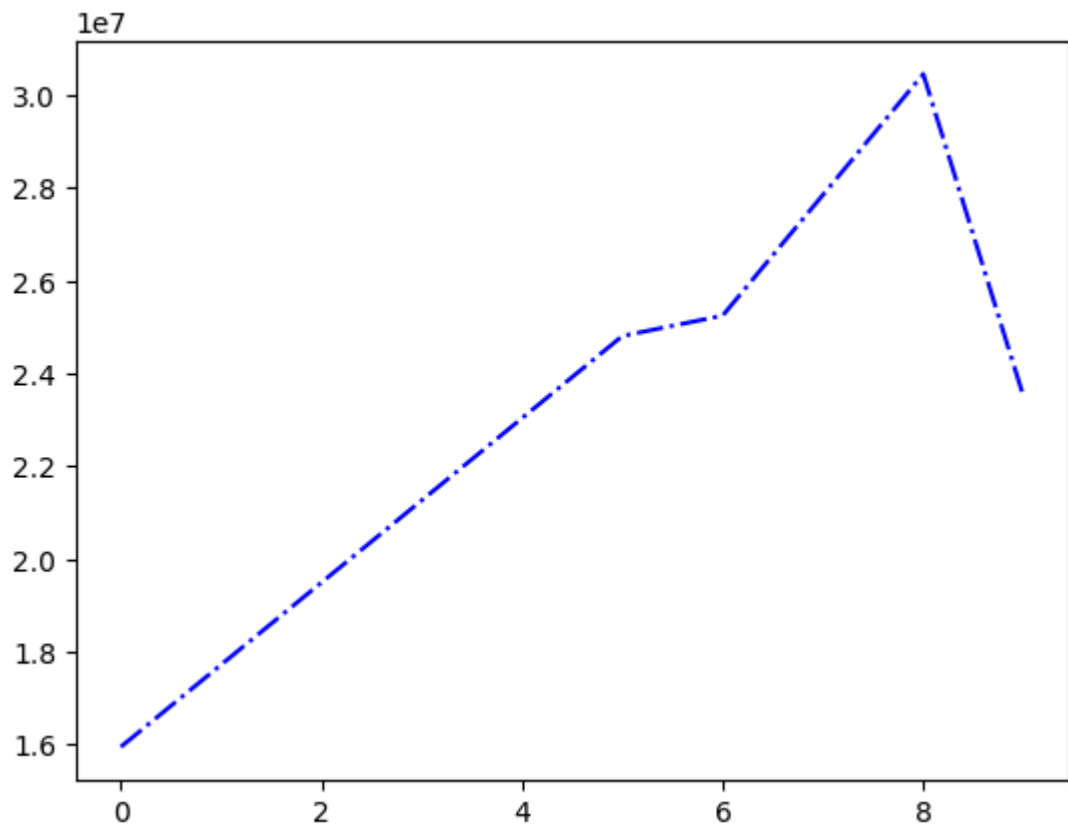
```
In [33]: plt.plot(Salary[0], c = 'b', ls='--')
```

```
Out[33]: [<matplotlib.lines.Line2D at 0x272cfa37750>]
```



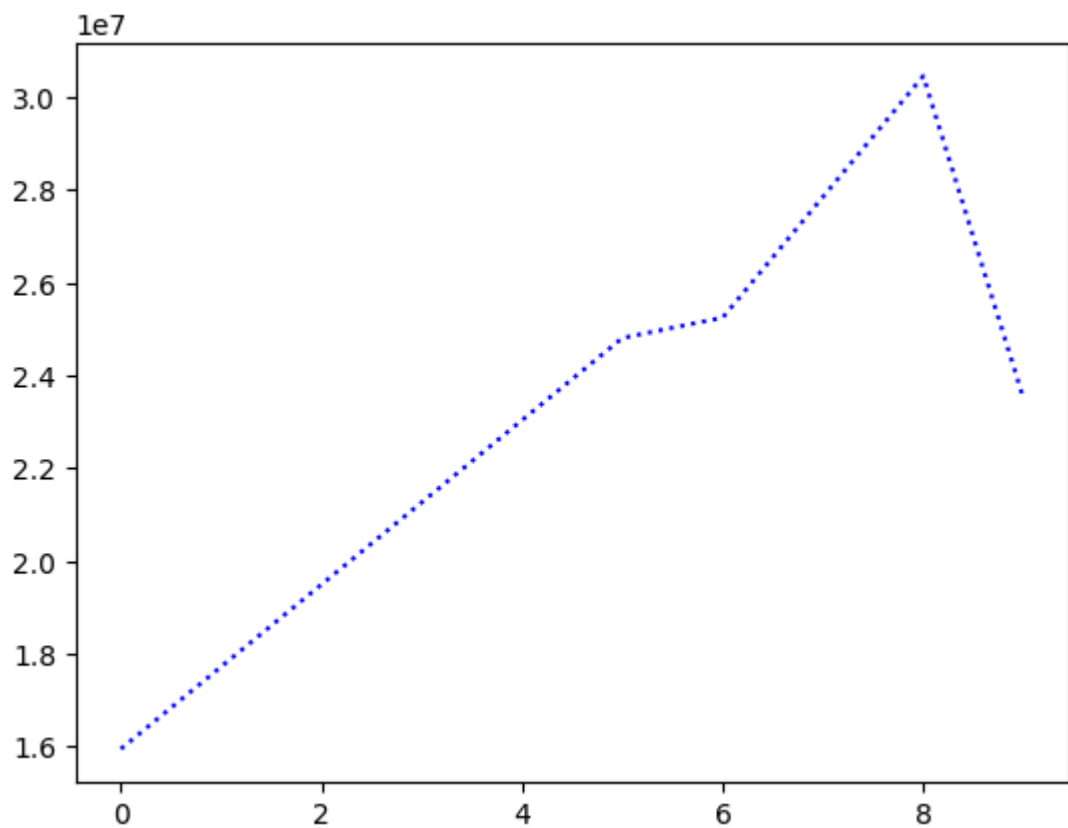
```
In [34]: plt.plot(Salary[0], c = 'b', ls='-.')
```

```
Out[34]: [<matplotlib.lines.Line2D at 0x272cfab5d10>]
```

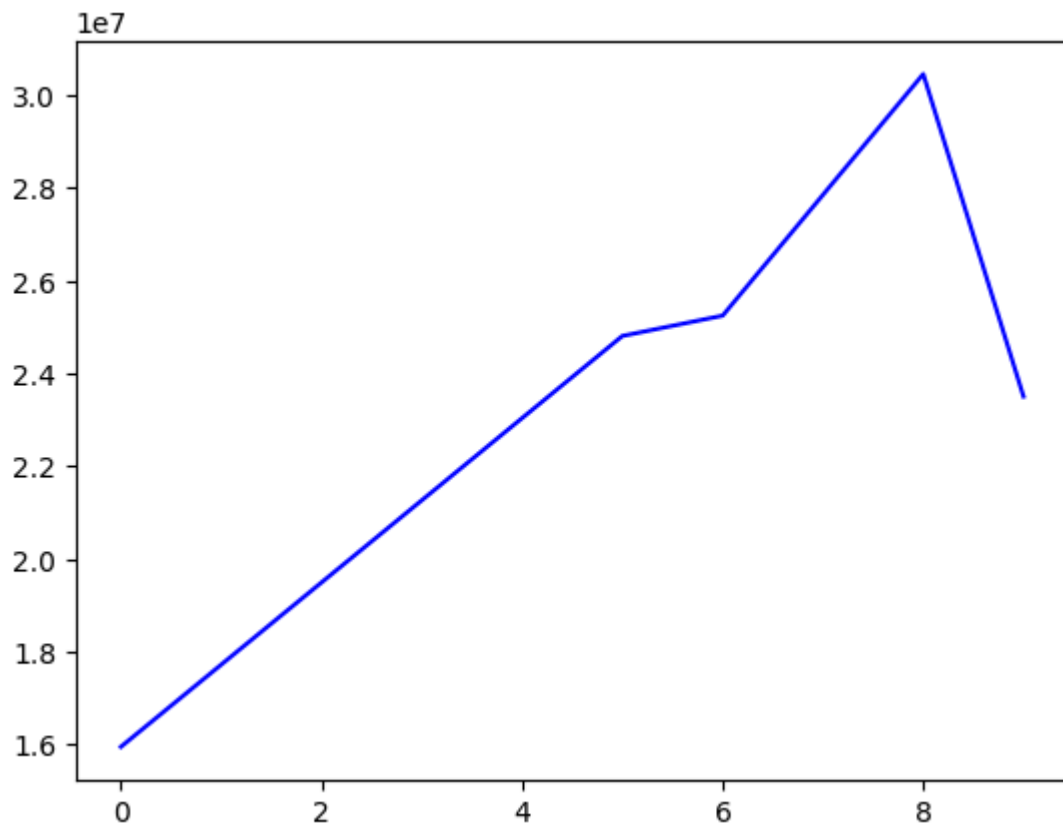
```
In [35]: plt.plot(Salary[0], c = 'b', ls='-.')
```

```
Out[35]: [<matplotlib.lines.Line2D at 0x272cfb402d0>]
```



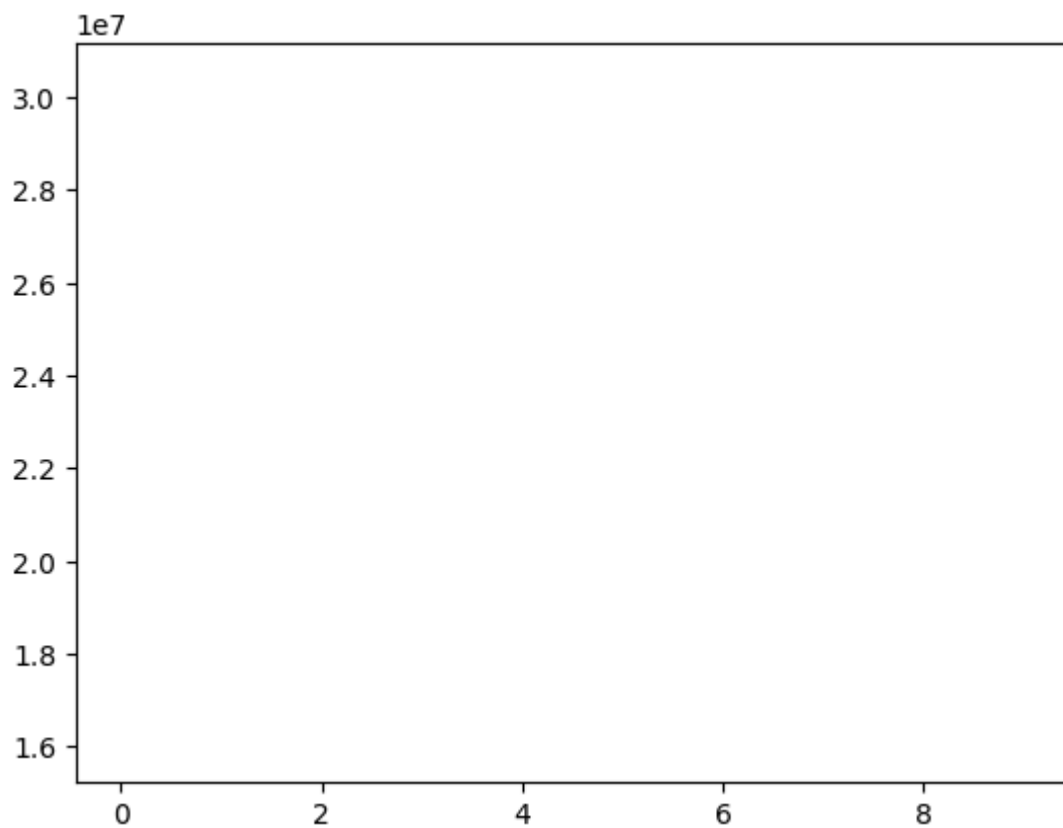
```
In [36]: plt.plot(Salary[0], c = 'b', ls='-.')
```

```
Out[36]: [<matplotlib.lines.Line2D at 0x272cfb9a850>]
```



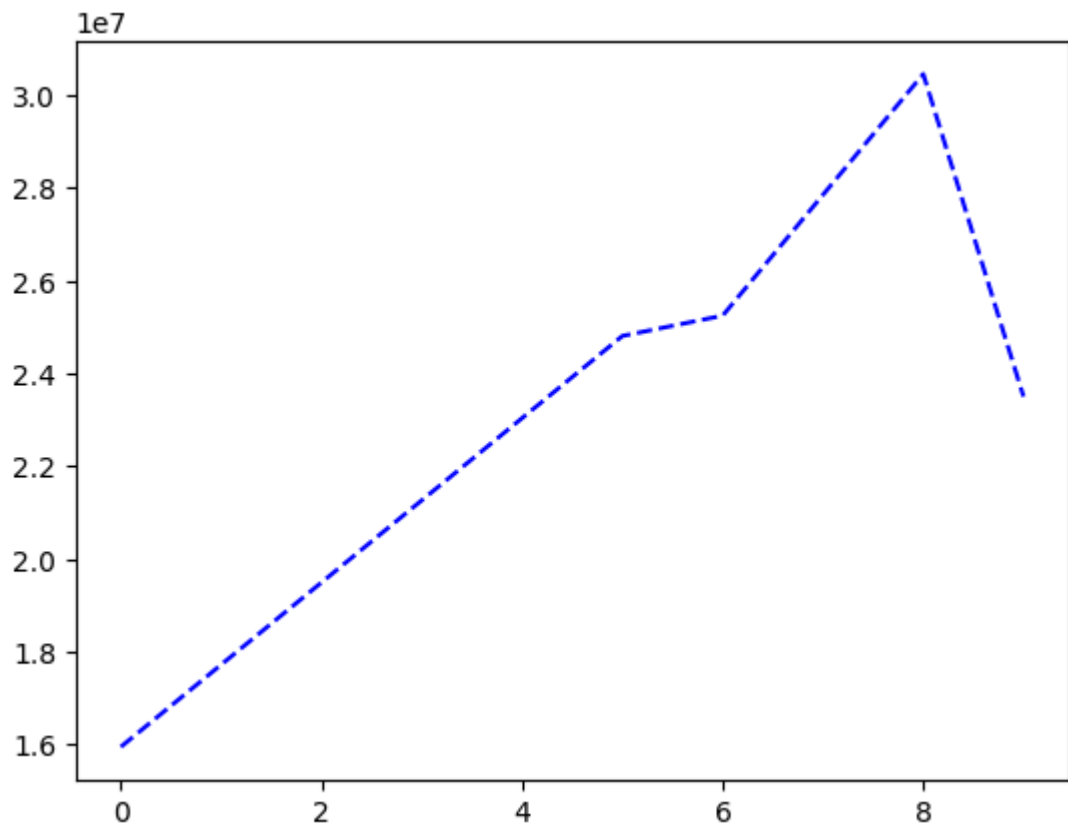
```
In [37]: plt.plot(Salary[0], c = 'b', ls='None')
```

```
Out[37]: [<matplotlib.lines.Line2D at 0x272cfc18e10>]
```



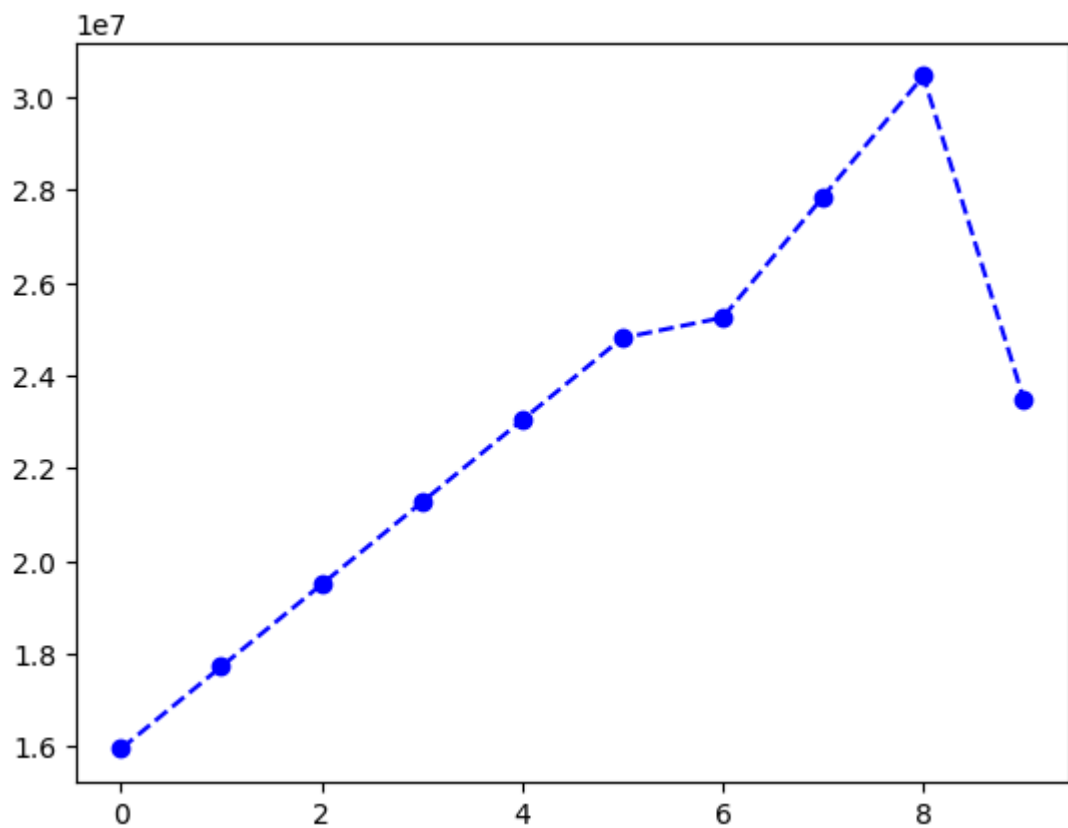
```
In [38]: plt.plot(Salary[0], c = 'b', ls='--')
```

```
Out[38]: [<matplotlib.lines.Line2D at 0x272cf93f390>]
```



```
In [39]: plt.plot(Salary[0], c = 'b', ls='--',marker = 'o') # Linear trend
```

```
Out[39]: [<matplotlib.lines.Line2D at 0x272cf9b9950>]
```



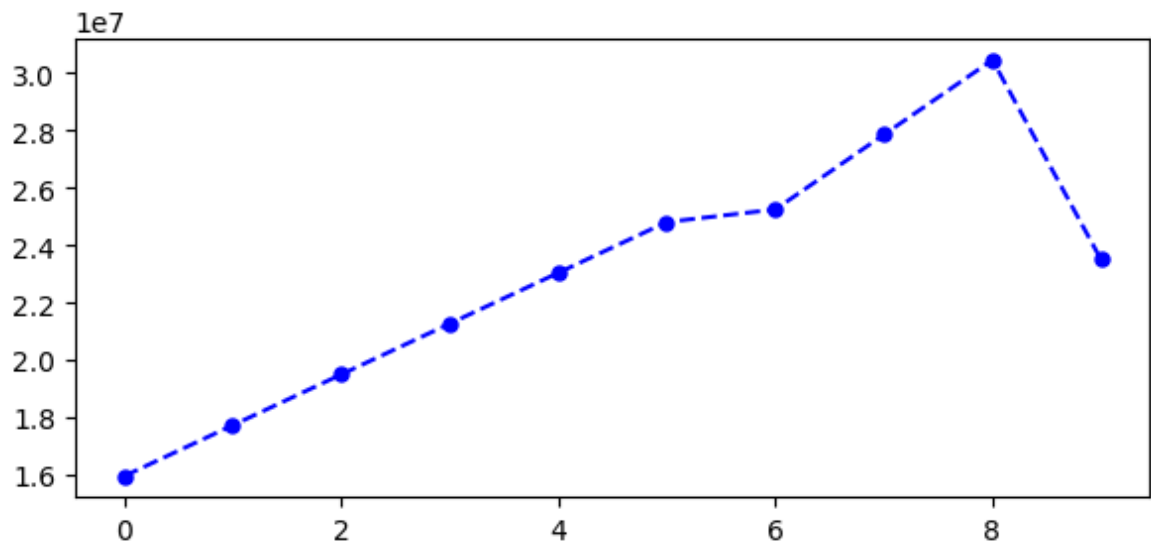
```
In [40]: Games[0]
```

```
Out[40]: array([80, 77, 82, 82, 73, 82, 58, 78, 6, 35])
```

```
In [41]: %matplotlib inline
plt.rcParams['figure.figsize'] = 7,3
```

```
In [42]: plt.plot(Salary[0], c = 'b', ls='--',marker = 'o',ms = 5) # linear trend
```

```
Out[42]: [<matplotlib.lines.Line2D at 0x272cfd7bb10>]
```



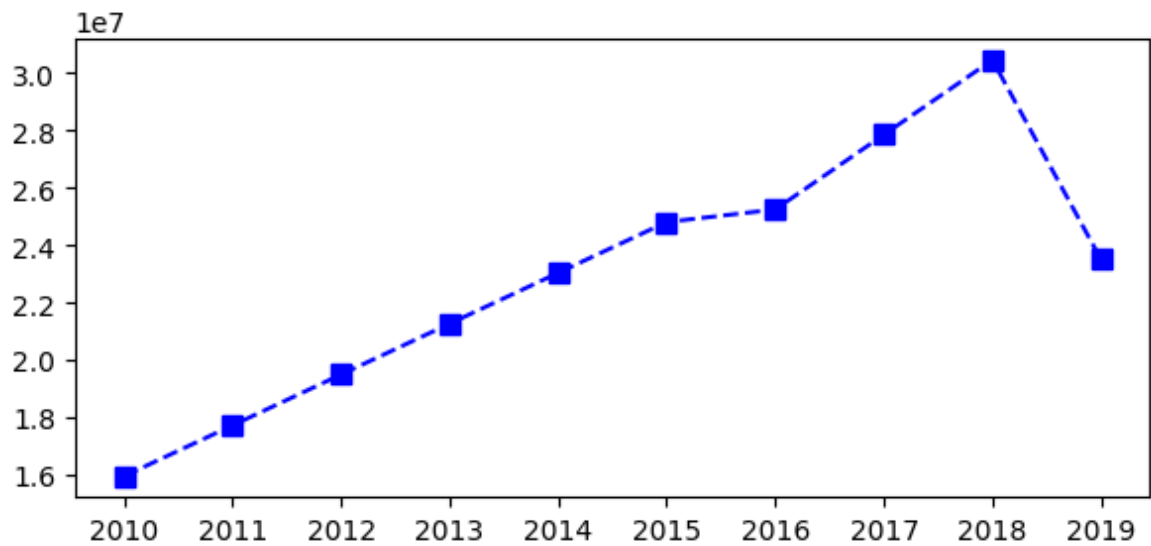
```
In [43]: Sdict
```

```
Out[43]: {'2010': 0,
          '2011': 1,
          '2012': 2,
          '2013': 3,
          '2014': 4,
          '2015': 5,
          '2016': 6,
          '2017': 7,
          '2018': 8,
          '2019': 9}
```

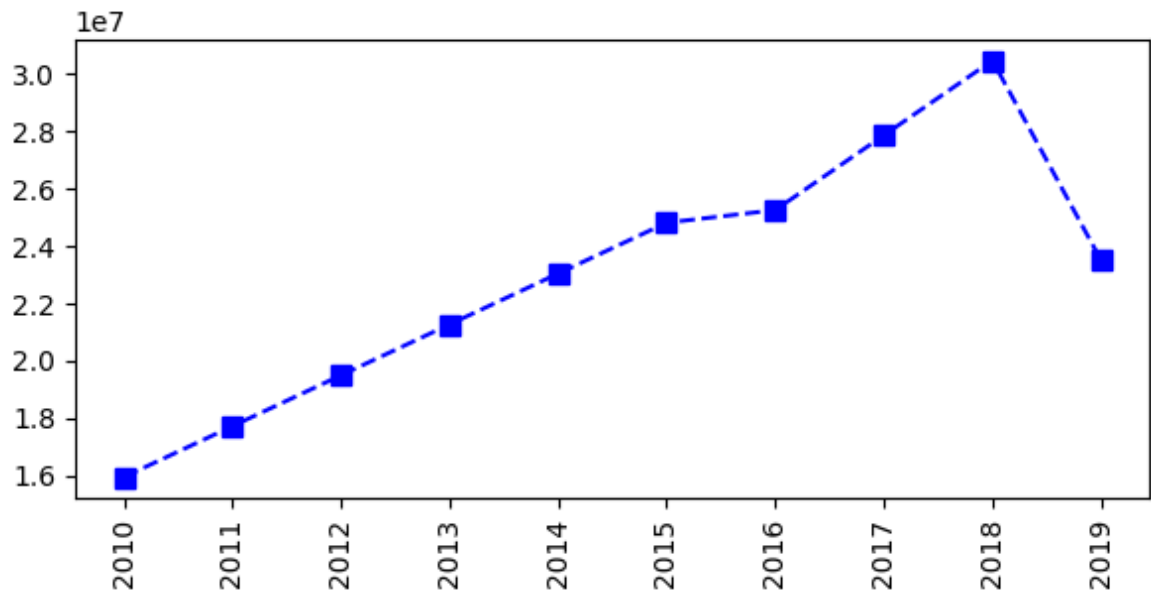
```
In [44]: Pdict
```

```
Out[44]: {'Sachin': 0,
          'Rahul': 1,
          'Smith': 2,
          'Sami': 3,
          'Pollard': 4,
          'Morris': 5,
          'Samson': 6,
          'Dhoni': 7,
          'Kohli': 8,
          'Sky': 9}
```

```
In [45]: plt.plot(Salary[0], c = 'b', ls='--',marker = 's',ms = 7) # linear trend
plt.xticks(list(range(0,10)),Seasons)
plt.show()
```



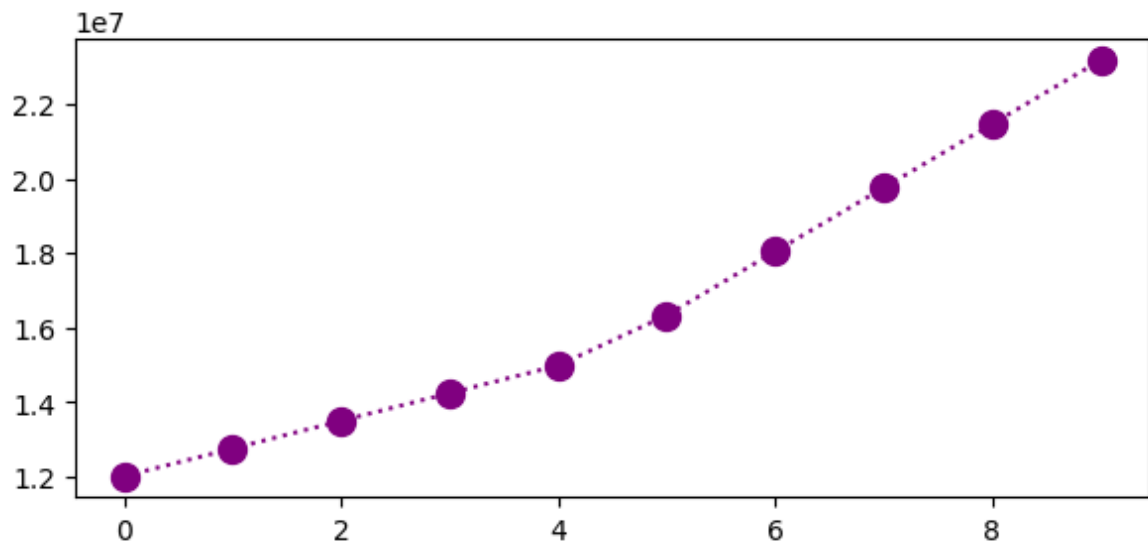
```
In [46]: plt.plot(Salary[0], c = 'b', ls='--',marker = 's',ms = 7) # linear trend
plt.xticks(list(range(0,10)),Seasons , rotation = 'vertical')
plt.show()
```



```
In [47]: Salary[1]
```

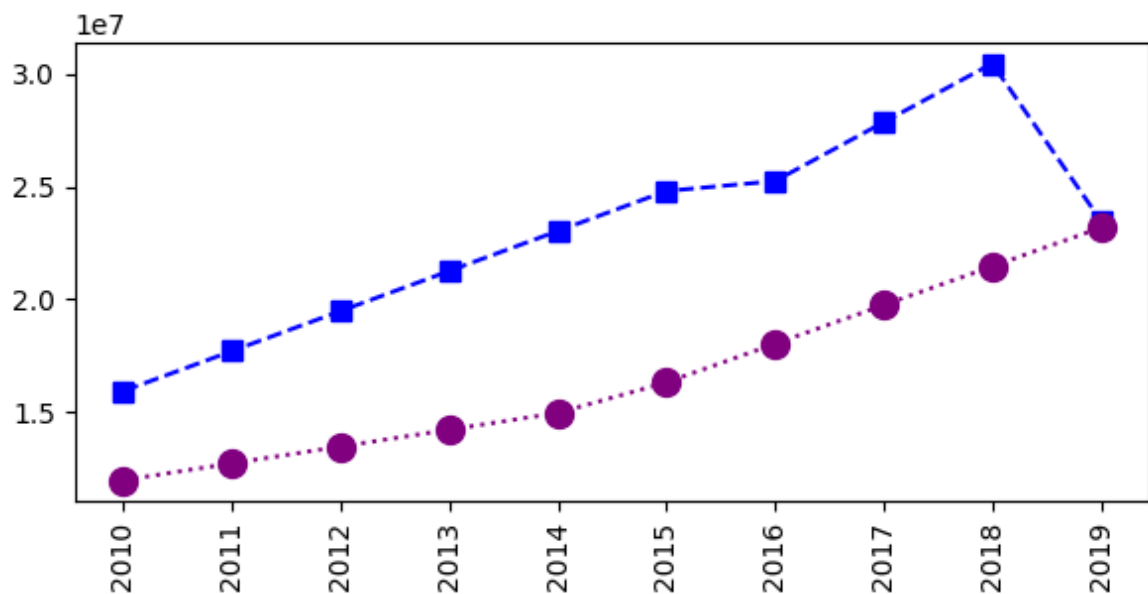
```
Out[47]: array([12000000, 12744189, 13488377, 14232567, 14976754, 16324500,
18038573, 19752645, 21466718, 23180790])
```

```
In [48]: plt.plot(Salary[1], c='purple', ls = ':', marker = 'o', ms = 10)
plt.show()
```

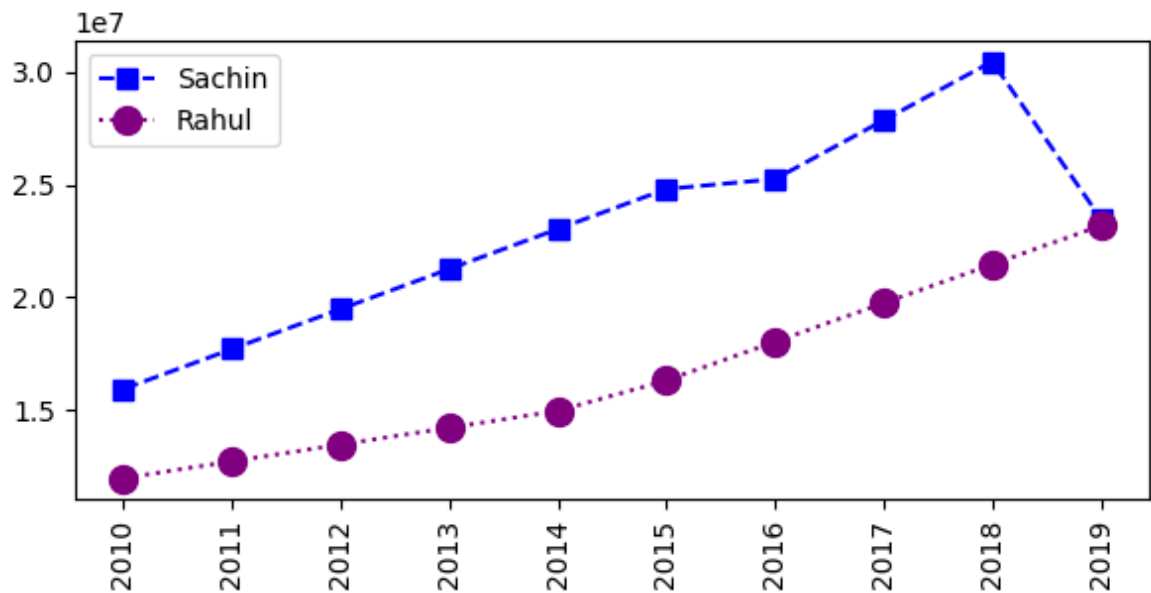


comparison

```
In [49]: plt.plot(Salary[0], c = 'b', ls='--',marker = 's',ms = 7) # linear trend
plt.xticks(list(range(0,10)),Seasons , rotation = 'vertical')
plt.plot(Salary[1], c='purple', ls = ':', marker = 'o', ms = 10, label = Players
plt.show()
```



```
In [50]: plt.plot(Salary[0], c = 'b', ls='--',marker = 's',ms = 7, label = Players[0]) #
plt.plot(Salary[1], c='purple', ls = ':', marker = 'o', ms = 10, label = Players
plt.xticks(list(range(0,10)),Seasons , rotation = 'vertical')
plt.legend()
plt.show()
```

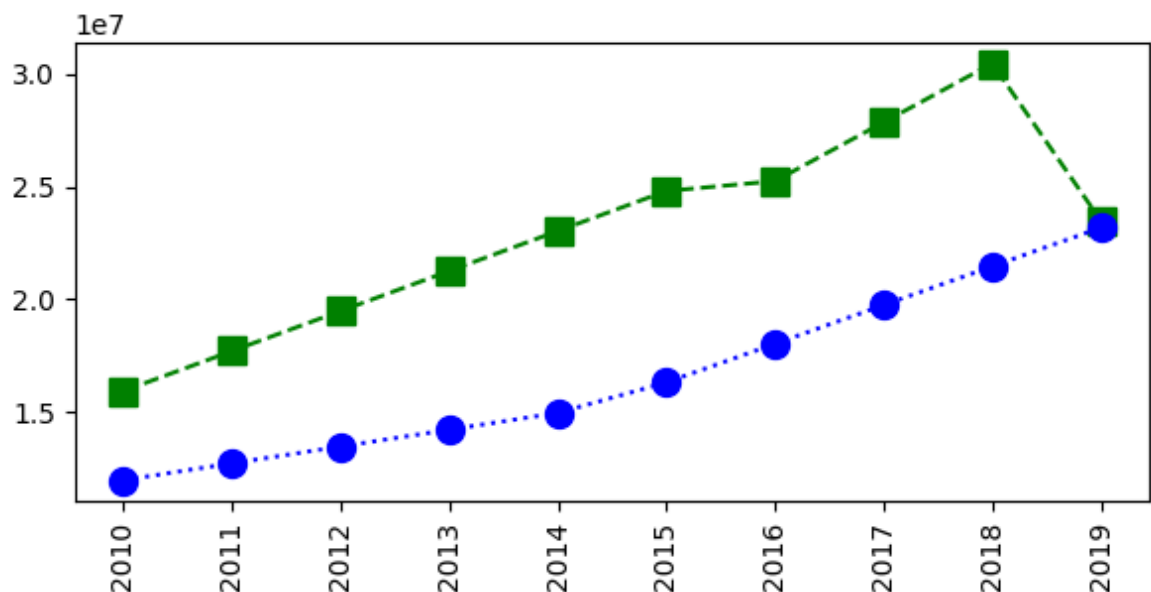


In [51]: `# # More visualization`

```
In [52]: plt.plot(Salary[0], c='Green', ls = '--', marker = 's', ms = 10, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = ':', marker = 'o', ms = 10, label = Players[1])

plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

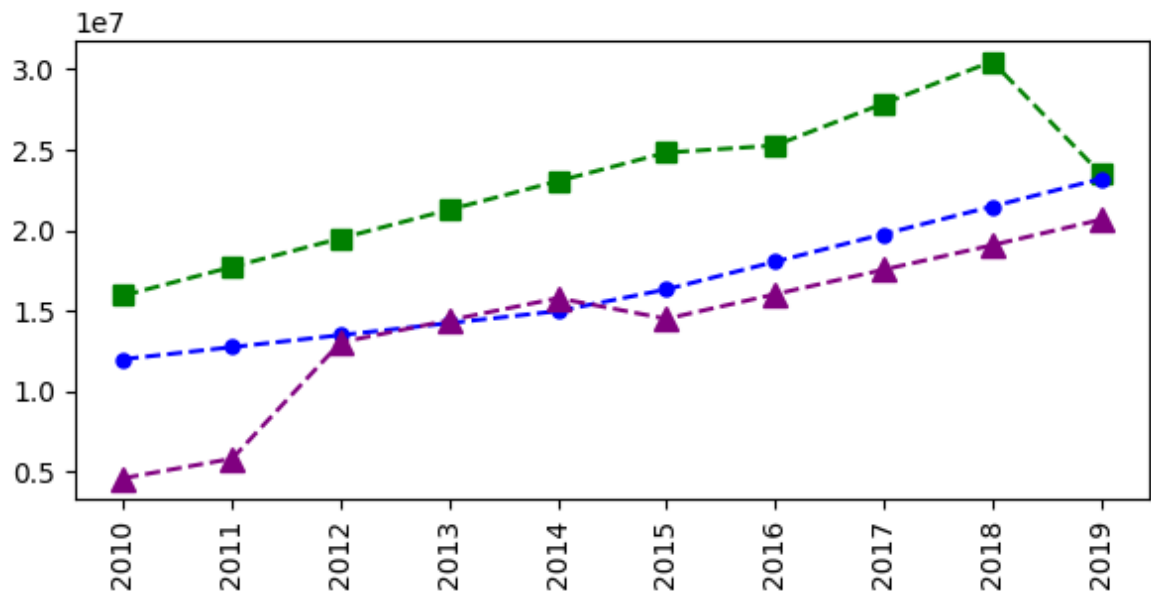
plt.show()
```



```
In [53]: plt.plot(Salary[0], c='Green', ls = '--', marker = 's', ms = 7, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = '--', marker = 'o', ms = 5, label = Players[1])
plt.plot(Salary[2], c='purple', ls = '--', marker = '^', ms = 8, label = Players[2])

plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

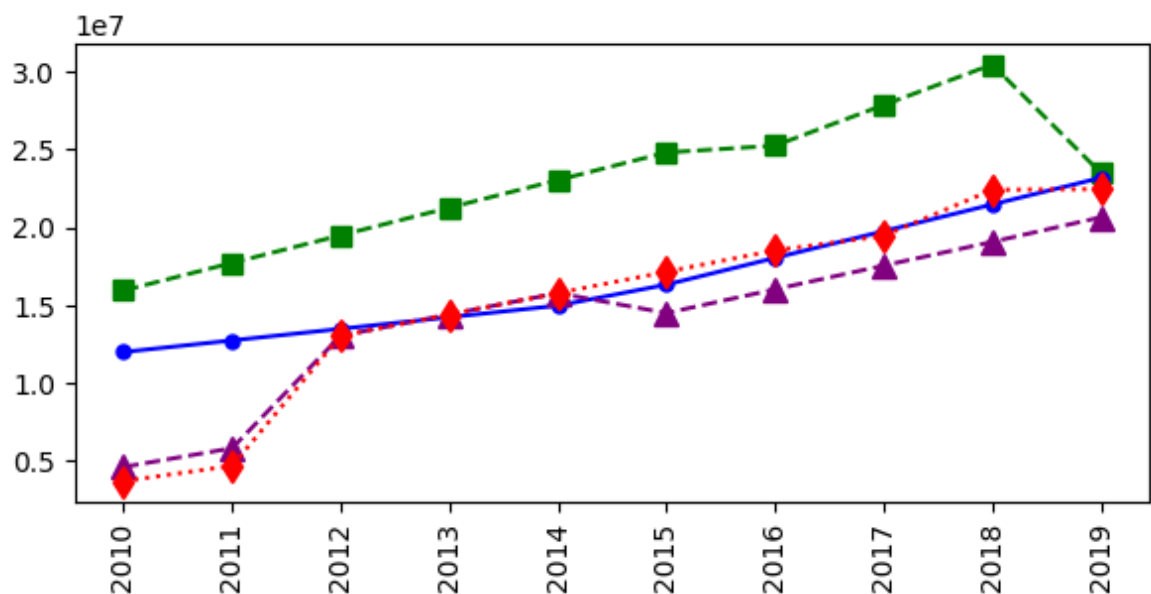
plt.show()
```



```
In [54]: plt.plot(Salary[0], c='Green', ls = '--', marker = 's', ms = 7, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = '-', marker = 'o', ms = 5, label = Players[1])
plt.plot(Salary[2], c='purple', ls = '--', marker = '^', ms = 8, label = Players[2])
plt.plot(Salary[3], c='Red', ls = ':', marker = 'd', ms = 8, label = Players[3])

plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

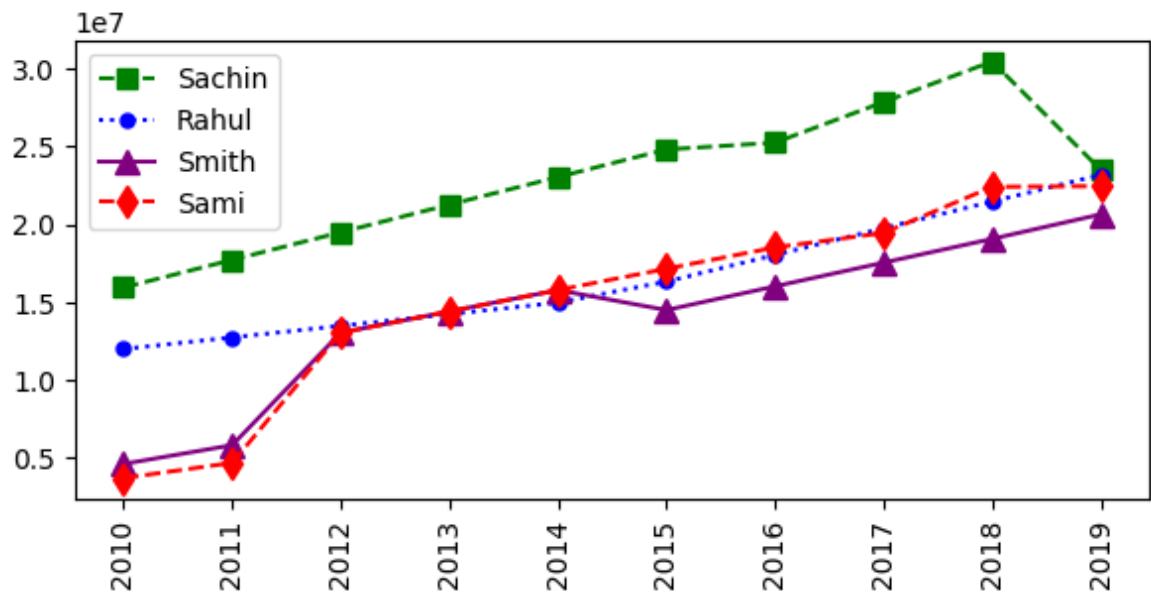
plt.show()
```



```
In [55]: # how to add legned in visualisation

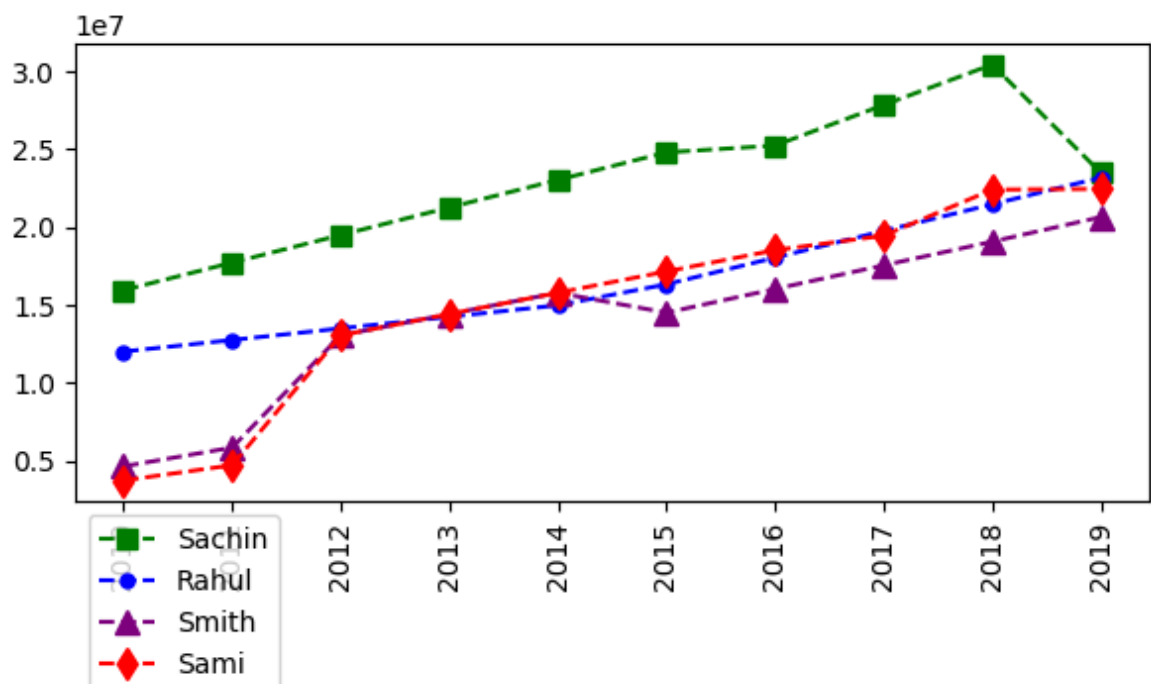
plt.plot(Salary[0], c='Green', ls = '--', marker = 's', ms = 7, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = '-', marker = 'o', ms = 5, label = Players[1])
plt.plot(Salary[2], c='purple', ls = '--', marker = '^', ms = 8, label = Players[2])
plt.plot(Salary[3], c='Red', ls = ':', marker = 'd', ms = 8, label = Players[3])
plt.legend()
plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

plt.show()
```

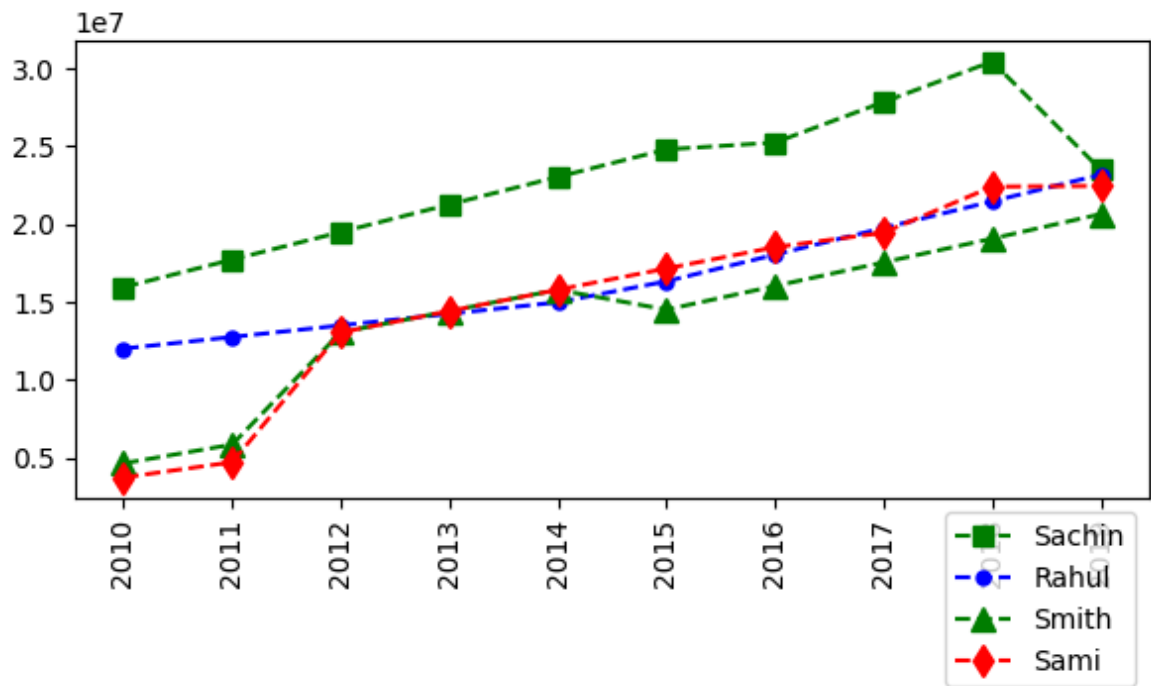
```
In [56]: plt.plot(Salary[0], c='Green', ls = '--', marker = 's', ms = 7, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = '--', marker = 'o', ms = 5, label = Players[1])
plt.plot(Salary[2], c='purple', ls = '--', marker = '^', ms = 8, label = Players[2])
plt.plot(Salary[3], c='Red', ls = '--', marker = 'd', ms = 8, label = Players[3])
plt.legend(loc = 'upper left',bbox_to_anchor=(0,0) )
plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

plt.show()
```



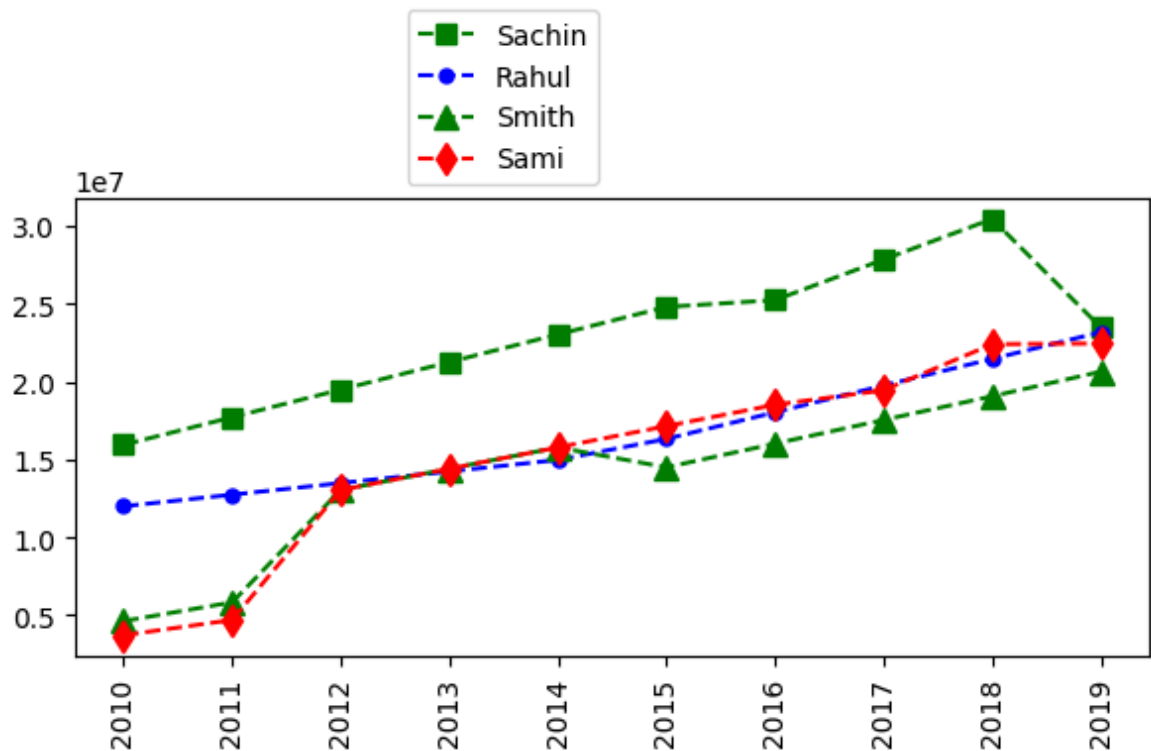
```
In [57]: plt.plot(Salary[0], c='Green', ls = '--', marker = 's', ms = 7, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = '--', marker = 'o', ms = 5, label = Players[1])
plt.plot(Salary[2], c='Green', ls = '--', marker = '^', ms = 8, label = Players[2])
plt.plot(Salary[3], c='Red', ls = '--', marker = 'd', ms = 8, label = Players[3])
plt.legend(loc = 'upper right',bbox_to_anchor=(1,0) )
plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

plt.show()
```



```
In [58]: plt.plot(Salary[0], c='Green', ls='--', marker='s', ms=7, label=Players[0])
plt.plot(Salary[1], c='Blue', ls='--', marker='o', ms=5, label=Players[1])
plt.plot(Salary[2], c='Green', ls='--', marker='^', ms=8, label=Players[2])
plt.plot(Salary[3], c='Red', ls='--', marker='d', ms=8, label=Players[3])
plt.legend(loc='lower right', bbox_to_anchor=(0.5, 1))
plt.xticks(list(range(0, 10)), Seasons, rotation='vertical')

plt.show()
```

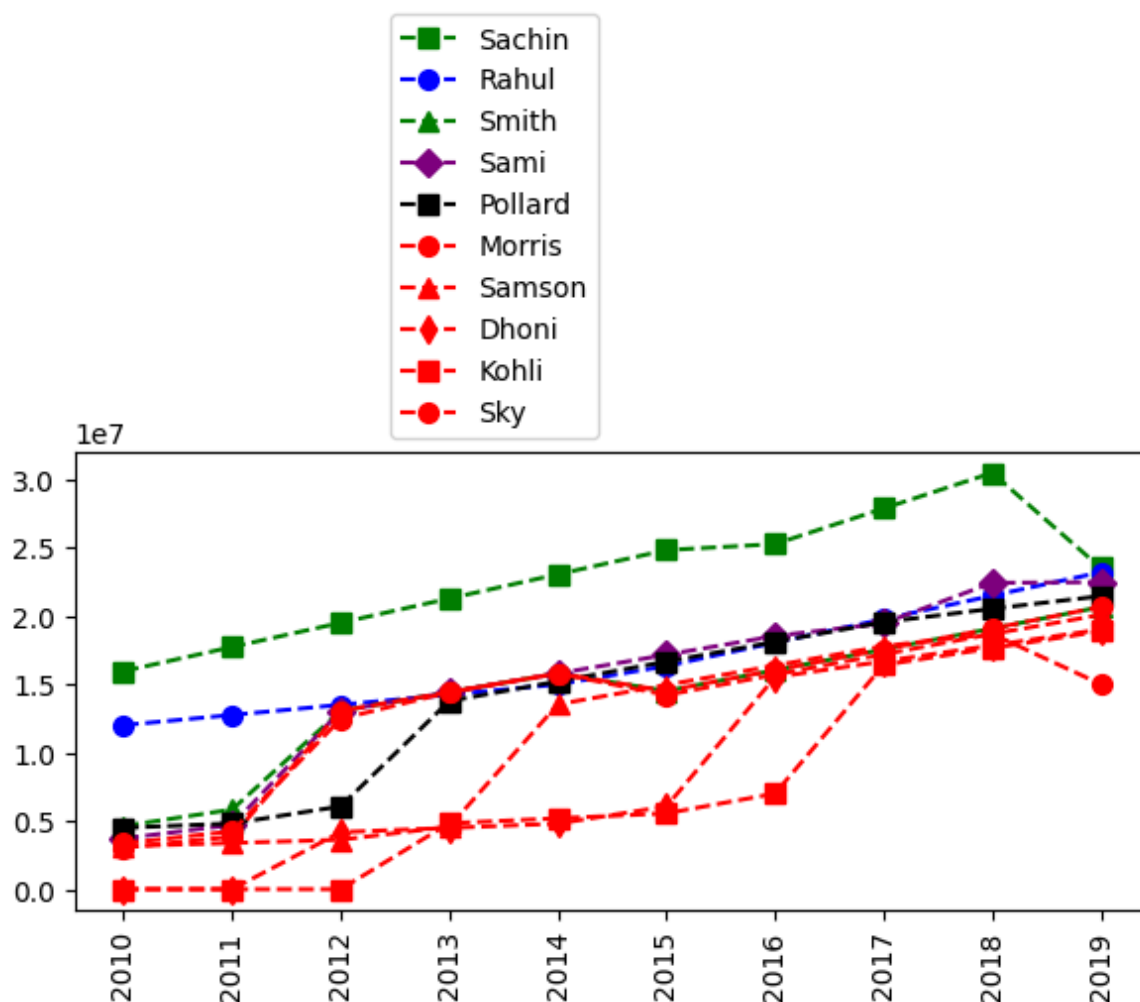


```
In [62]: plt.plot(Salary[0], c='Green', ls='--', marker='s', ms=7, label=Players[0])
plt.plot(Salary[1], c='Blue', ls='--', marker='o', ms=7, label=Players[1])
plt.plot(Salary[2], c='Green', ls='--', marker='^', ms=7, label=Players[2])
plt.plot(Salary[3], c='Purple', ls='--', marker='D', ms=7, label=Players[3])
plt.plot(Salary[4], c='Black', ls='--', marker='s', ms=7, label=Players[4])
```

```
plt.plot(Salary[5], c='Red', ls = '--', marker = 'o', ms = 7, label = Players[5])
plt.plot(Salary[6], c='Red', ls = '--', marker = '^', ms = 7, label = Players[6])
plt.plot(Salary[7], c='Red', ls = '--', marker = 'd', ms = 7, label = Players[7])
plt.plot(Salary[8], c='Red', ls = '--', marker = 's', ms = 7, label = Players[8])
plt.plot(Salary[9], c='Red', ls = '--', marker = 'o', ms = 7, label = Players[9])

plt.legend(loc = 'lower right',bbox_to_anchor=(0.5,1) )
plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

plt.show()
```

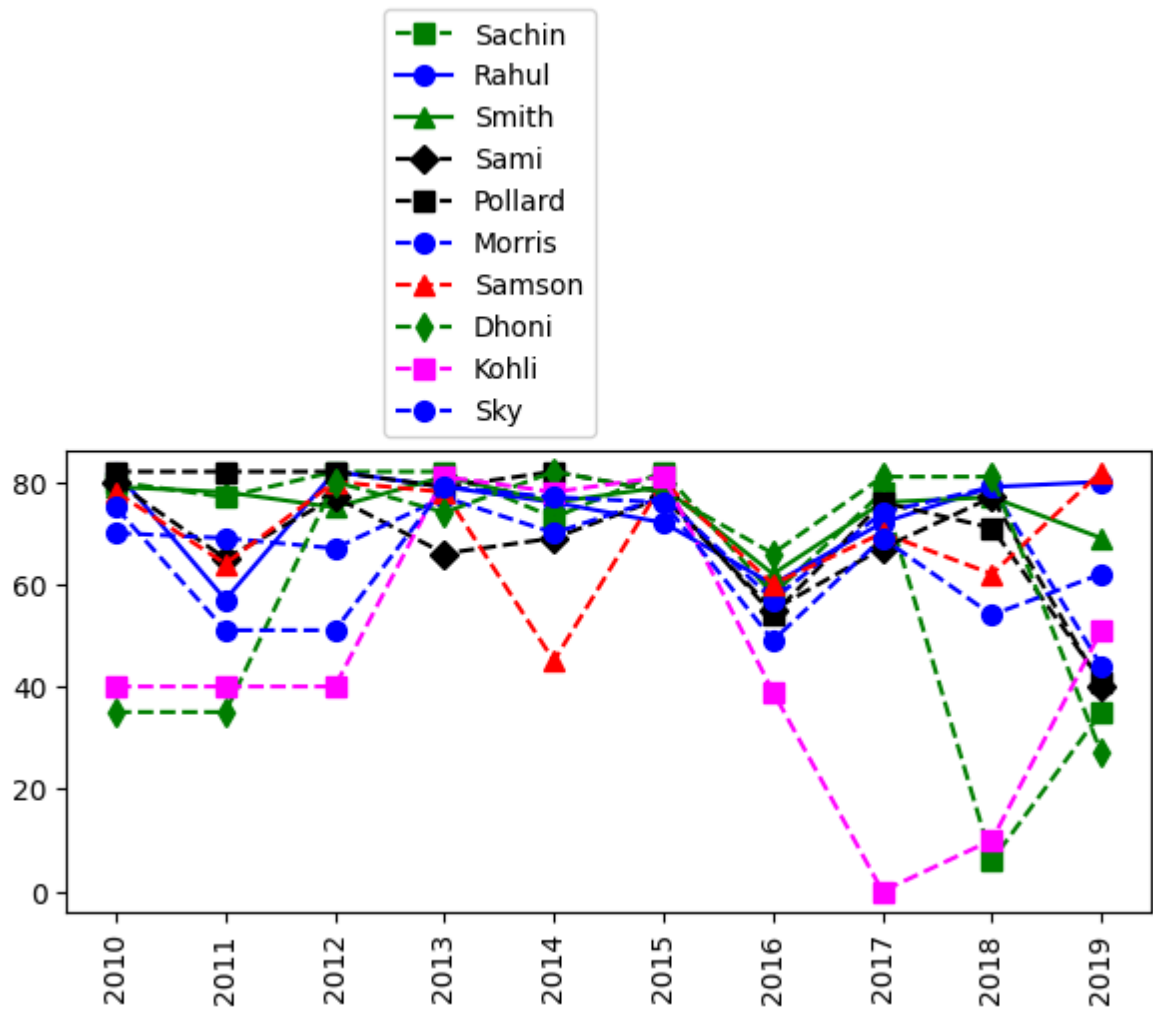


In [61]: *# we can visualize the how many games played by a player*

```
plt.plot(Games[0], c='Green', ls = '--', marker = 's', ms = 7, label = Players[0])
plt.plot(Games[1], c='Blue', ls = '--', marker = 'o', ms = 7, label = Players[1])
plt.plot(Games[2], c='Green', ls = '--', marker = '^', ms = 7, label = Players[2])
plt.plot(Games[3], c='Black', ls = '--', marker = 'D', ms = 7, label = Players[3])
plt.plot(Games[4], c='Black', ls = '--', marker = 's', ms = 7, label = Players[4])
plt.plot(Games[5], c='Blue', ls = '--', marker = 'o', ms = 7, label = Players[5])
plt.plot(Games[6], c='red', ls = '--', marker = '^', ms = 7, label = Players[6])
plt.plot(Games[7], c='Green', ls = '--', marker = 'd', ms = 7, label = Players[7])
plt.plot(Games[8], c='Magenta', ls = '--', marker = 's', ms = 7, label = Players[8])
plt.plot(Games[9], c='Blue', ls = '--', marker = 'o', ms = 7, label = Players[9])

plt.legend(loc = 'lower right',bbox_to_anchor=(0.5,1) )
plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

plt.show()
```



In []: